WORKED SOLUTION

STEP I 1996, Question 2

Pure • Beta Integrals

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q2

The key idea

The integrands in (i) and (ii) are the same function written two ways: $\alpha x + (1-x) = 1 + (\alpha - 1)x$. So one integral, worked out in two ways, gives two descriptions of the same polynomial in α. Part (i) gives a closed form, part (ii) gives the coefficients as integrals, and comparing coefficients does the rest. No integration by parts is needed.

Solution

Part (i)

Substitute $u = 1 + (\alpha - 1)x$, so $\mathrm{d}u = (\alpha - 1)\,\mathrm{d}x$; when $x = 0$, $u = 1$ and when $x = 1$, $u = \alpha$. Since $\alpha \neq 1$ we may divide by $\alpha - 1$:

$$\int_0^1 (1 + (\alpha - 1)x)^n \,\mathrm{d}x = \frac{1}{\alpha - 1}\int_1^\alpha u^n \,\mathrm{d}u = \frac{1}{\alpha - 1}\left[\frac{u^{n+1}}{n+1}\right]_1^\alpha = \frac{\alpha^{n+1} - 1}{(n+1)(\alpha - 1)},$$

as required. (The substitution is fine even if $\alpha < 1$ or $\alpha < 0$: it is linear, and the formula $\int_1^\alpha u^n \,\mathrm{d}u$ holds with the limits in either order. Equivalently, just differentiate $(1 + (\alpha - 1)x)^{n+1}/((n+1)(\alpha - 1))$.)

Part (ii)

Expand the integrand by the binomial theorem, treating αx and $1 - x$ as the two terms:

$$(\alpha x + (1-x))^n = \sum_{k=0}^{n} \binom{n}{k} (\alpha x)^k (1-x)^{n-k} = \sum_{k=0}^{n} \alpha^k \binom{n}{k} x^k (1-x)^{n-k}.$$

This is a finite sum, so we may integrate term by term; α is a constant as far as the integration over x is concerned:

$$\int_0^1 (\alpha x + (1-x))^n \,\mathrm{d}x = \sum_{k=0}^{n} \alpha^k \left[\binom{n}{k} \int_0^1 x^k (1-x)^{n-k} \,\mathrm{d}x\right].$$

Each bracket is a number not depending on α, so this is a polynomial in α of degree at most n, and the coefficient of α^k for $0 \le k \le n$ is $\binom{n}{k}\int_0^1 x^k(1-x)^{n-k}\,\mathrm{d}x$, as required.

Part (iii)

Write $P(\alpha)$ for the polynomial in (ii). Because $\alpha x + (1-x) = 1 + (\alpha-1)x$, part (i) says that for every $\alpha \neq 1$,

$$P(\alpha) = \frac{\alpha^{n+1}-1}{(n+1)(\alpha-1)} = \frac{1}{n+1}(1+\alpha+\alpha^2+\cdots+\alpha^n),$$

using the geometric series factorisation $\alpha^{n+1}-1 = (\alpha-1)(1+\alpha+\cdots+\alpha^n)$.

Ask yourself We only know these two polynomials agree for $\alpha \neq 1$. Why may we compare coefficients?

The difference of the two sides is a polynomial of degree at most n that vanishes at every $\alpha \neq 1$, so it has infinitely many roots. A nonzero polynomial of degree at most n has at most n roots, so the difference is the zero polynomial: every coefficient matches. (Alternatively, both sides are continuous, so they also agree at $\alpha = 1$.)

On the right, every coefficient of α^k, $0 \le k \le n$, equals $\frac{1}{n+1}$. Comparing with (ii),

$$\binom{n}{k}\int_0^1 x^k(1-x)^{n-k}\,\mathrm{d}x = \frac{1}{n+1},$$

and so

$$\int_0^1 x^k(1-x)^{n-k}\,\mathrm{d}x = \frac{1}{(n+1)\binom{n}{k}} = \frac{k!\,(n-k)!}{(n+1)\,n!} = \frac{k!\,(n-k)!}{(n+1)!},$$

which is the required result.

Checks

- $k=0$: $\int_0^1 (1-x)^n\,\mathrm{d}x = \frac{1}{n+1}$, and the formula gives $\frac{n!}{(n+1)!} = \frac{1}{n+1}$.
- $n=2$, $k=1$: $\int_0^1 (x-x^2)\,\mathrm{d}x = \frac{1}{2}-\frac{1}{3} = \frac{1}{6} = \frac{1!\,1!}{3!}$.
- Numerically, $\int_0^1 x^2(1-x)^3\,\mathrm{d}x \approx 0.016667 = \frac{1}{60} = \frac{2!\,3!}{6!}$; we checked (i) by Simpson's rule for random α and n, and (iii) exactly for all $0 \le k \le n \le 12$.
- A pleasing consequence: the coefficients of P are all equal, so $\int_0^1 \binom{n}{k}x^k(1-x)^{n-k}\,\mathrm{d}x$ does not depend on k.

Common slips

- Comparing coefficients without remarking that (i) was only proved for $\alpha \neq 1$; one sentence about polynomials (or continuity) closes the gap.

- Expanding $(1+(\alpha-1)x)^n$ in powers of $(\alpha-1)$ in (ii): that gives the right polynomial but not in powers of α. Expand $\big(\alpha x+(1-x)\big)^n$ instead.

Beyond STEP

The integral $\int_0^1 x^{a-1}(1-x)^{b-1}\,\mathrm{d}x$ is Euler's *beta function* $B(a,b)$, and the result here is the integer case of $B(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$. In probability, it says that if p is chosen uniformly from $[0,1]$ and then n coins with head-probability p are tossed, every number of heads from 0 to n is equally likely, each with probability $\frac{1}{n+1}$ (this is exactly what "all coefficients equal" means), the starting point of Bayes's own famous essay. Grimmett and Welsh, *Probability: An Introduction*, develops the beta and gamma distributions behind this.

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